Welcome to the Data Science Specialist Curriculum

This curriculum is designed to take aspiring data professionals from the fundamentals of programming and database operations to production-level machine learning and experimentation. It emphasizes the ability to translate open-ended business problems into structured analytical tasks, covering: Business Problem → Data → EDA → Statistical Inference → Modeling → Metric Evaluation → Business Insights → Deployment & Stakeholder Communication.

Table of Stages / Content Map:

- Stage 8 — Experimentation & A/B Testing (Sample sizing, power, multiple testing)
- Stage 9 — Business & Product Analytics (KPIs, funnels, cohort analysis, CLV)
- Stage 10 — Time Series Forecasting (Stationarity, Autocorrelations, ARIMA/SARIMA)
- Stage 11 — Advanced Data Science (Causal inference, Survival analysis, Bayesian stats)
- Stage 12 — Data Science Engineering (ETL, APIs, FastAPI, Docker, MLflow, Monitoring)
- Stage 13 — 20 Progression Projects (5 Beginner, 5 Inter., 5 Adv., 5 Business/Product)
- Stage 14 — Data Science Interview Preparation & Case Studies

Stage 0 — Prerequisites

Establish a strong base in tools, languages, and libraries before diving into statistical modelling.

Python

[MUST KNOW] Core language used in modern data science. Focus on functions, loops, lists, dicts, and file operations.

NumPy

[MUST KNOW] Numerical computing. Focus on array vectorization, slicing, array transformations, and dot products.

Pandas

[MUST KNOW] Dataframes, cleaning operations, filtering rows, grouping, pivoting, merging, datetime conversions.

Matplotlib & Seaborn

[MUST KNOW] Visualizing feature relationships, distributions, histograms, heatmaps, box plots.

SQL

[MUST KNOW] Retrieving, aggregating, and joining data from relational database systems.

Git & GitHub

[SHOULD KNOW] Version control, branching, commits, pulls, and pushing code to remote repositories.

Jupyter Notebooks

[MUST KNOW] Interactive programming environment for coding, testing, and visualizing EDA code.

Stage 1 — Mathematics & Statistics

Statistical thinking is the core value driver of a data scientist. Master descriptive and inferential statistics.

Descriptive Statistics

Central Tendency (Mean, Median, Mode)

[MUST KNOW] Mean: average (sensitive to outliers). Median: midpoint (outlier robust). Mode: most frequent.

Spread (Variance, Std Dev, Percentiles, IQR)

[MUST KNOW] Variance & Std Dev: measure dispersion. Percentiles: value below which a percentage of observations fall. IQR: Q3 - Q1 (outlier detection bounds).

Shape (Skewness, Kurtosis)

[SHOULD KNOW] Skewness: measures symmetry of distribution. Kurtosis: measures tailedness/peakedness (platykurtic vs. leptokurtic).

Probability & Distributions

Conditional Probability & Bayes Theorem

[MUST KNOW] Bayes Theorem calculates post-event likelihood: $P(A|B) = P(B|A)*P(A) / P(B)$. Core of Bayesian updates and diagnostic tests.

Common Distributions

[MUST KNOW] Bernoulli/Binomial (binary events), Poisson (counts per time interval), Uniform (equal probabilities), Normal (Gaussian bells), Exponential (time between events).

Central Limit Theorem (CLT)

[MUST KNOW] Crucial theorem: The distribution of sample means approaches normal as sample size N grows, regardless of parent distribution shape.

Inferential Statistics & Hypothesis Testing

Drawing conclusions about populations from sample observations under probabilistic limits.

Confidence Intervals

[MUST KNOW] Range estimating a population parameter with a confidence level (e.g. 95%). Width decays as sample size increases.

Hypothesis Testing & p-Values

[MUST KNOW] Null Hypothesis (H0) vs. Alternative (H1). p-value: probability of observing sample statistics as extreme as collected, assuming H0 is true.

Errors & Power (Type I, Type II)

[MUST KNOW] Type I error (α): rejecting H0 when true (false positive). Type II error (β): retaining H0 when false (false negative). Statistical Power (1-β): probability of correctly rejecting a false H0.

Statistical Tests Guide

Test Name	When to Use	Assumptions	Common Pitfall
t-test (1-sample, 2-sample, paired)	Compare means of 1 or 2 groups when population variance is unknown.	Normal distribution of groups, equal variances (independent t-test).	Using it on highly skewed, small datasets without non-parametric backups.
ANOVA (One-Way / Two-Way)	Compare means across 3 or more independent groups.	Normality, homogeneity of variances, independence.	Running multiple t-tests instead of ANOVA, inflating Type I error rate.
Chi-Square Test of Independence	Compare relationships between 2 categorical variables.	Expected counts in each cell ≥ 5, independent observations.	Applying to continuous variables or small sample sizes.
Mann-Whitney U Test	Compare distributions of 2 groups when normality fails (non-parametric t-test).	Ordinal or continuous data, independent samples.	Interpreting it as a direct comparison of means (it compares ranks).
Kruskal-Wallis Test	Compare distributions across 3+ groups when normality fails (non-parametric ANOVA).	Ordinal or continuous data, independent samples.	Failing to perform post-hoc Dunn's tests to locate which groups differ.

Stage 2 — SQL

The language of database queries. Essential for extracting and processing structured datasets.

Aggregations & Grouping

[MUST KNOW] GROUP BY, HAVING, and aggregations (COUNT, SUM, AVG, MIN, MAX). Remember: HAVING filters aggregated groups; WHERE filters raw rows.

Subqueries & CTEs

[MUST KNOW] Common Table Expressions (WITH queries) and nested subqueries. CTEs are preferred for readability and optimization structures.

Window Functions & Ranking

[MUST KNOW] ROW_NUMBER(), RANK(), DENSE_RANK(), and partitions (`OVER(PARTITION BY ... ORDER BY ...)`). Crucial for filtering Top N entries per group.

LEAD & LAG operations

[MUST KNOW] Retrieving values from preceding (LAG) or succeeding (LEAD) rows. Crucial for calculating session durations and step conversion differences.

Query Optimization

[SHOULD KNOW] Reducing read times using index joins, avoiding SELECT *, avoiding Cartesian joins, and utilizing CTE boundaries effectively.

Advanced SQL Interview Problem Example

Problem: Given a table of user log-ins `user_logins (user_id, login_time)`, find users who logged in 3 or more consecutive days.

```
WITH unique_dates AS (
    SELECT DISTINCT user_id, CAST(login_time AS DATE) AS login_date
    FROM user_logins
),
ranked_dates AS (
    SELECT user_id, login_date,
           ROW_NUMBER() OVER(PARTITION BY user_id ORDER BY login_date) AS rn
    FROM unique_dates
),
grouped_dates AS (
    SELECT user_id,
           login_date - INTERVAL '1 day' * rn AS group_date
    FROM ranked_dates
)
SELECT user_id
FROM grouped_dates
GROUP BY user_id, group_date
HAVING COUNT(*) >= 3;
```

Stage 3 — Data Cleaning

Garbage in, garbage out. Cleanse data to ensure modeling pipelines receive high-quality inputs.

Missing Data Types

[MUST KNOW] 1) MCAR (Missing Completely at Random): missingness independent of variables. Impute safely. 2) MAR (Missing at Random): missingness depends on observed features. Impute using conditional rules. 3) MNAR (Missing Not at Random): missingness depends on unobserved values. Requires structural additions or indicator columns.

Outlier Treatment

[MUST KNOW] Detecting outliers via IQR (clipping to [Q1 - 1.5*IQR, Q3 + 1.5*IQR]) or Z-score boundaries. Avoid dropping outliers blindly; clip (winsorization) or apply transforms first.

Inconsistent Categories & Types

[MUST KNOW] Resolving mixed classifications (e.g. 'US', 'USA', 'United States' to 'USA') and casting column types (e.g. object to float or datetime).

Data Quality & Leakage validation

[MUST KNOW] Spotting duplicates, verifying input ranges, and identifying target variables data leakage (dropping features containing future target information).

Stage 4 — Exploratory Data Analysis (EDA)

Perform univariate, bivariate, and multivariate analysis to identify distribution shapes, target relationships, and anomalies.

Univariate Analysis

[MUST KNOW] Analyzing single columns. Plot histograms to inspect skewness, box plots for outliers, and value counts for categories.

Bivariate & Multivariate Analysis

[MUST KNOW] Analyzing two or more columns simultaneously. Use scatter plots to trace feature relationships, heatmaps for correlation coefficients, and pairplots for overall grids.

Systematic Dataset Approach

[MUST KNOW] When given an unfamiliar dataset: 1) Check dimensions (.shape) and column data types, 2) Calculate descriptive summaries (.describe()), 3) Scan null counts (.isnull().sum()), 4) Plot target variable distribution, 5) Compute correlation heatmap with target.

Stage 5 — Feature Engineering

Transforming features to improve model capacity and linear alignments.

Numerical Transformations

[MUST KNOW] Log transformations (`np.log1p`) to reduce heavy right skewness. Power transforms (Box-Cox, Yeo-Johnson) to make distributions Gaussian.

Feature Scaling

[MUST KNOW] StandardScaler (mean=0, std=1) for normal-behaving parameters. MinMaxScaler (scale to 0-1) for neural networks and distance algorithms. RobustScaler for outliers.

Categorical Encoding

[MUST KNOW] One-hot encoding for low-cardinality unordered fields. Target/Mean encoding for high-cardinality fields (e.g. zip codes) to prevent dimension explosions.

Feature Selection

[SHOULD KNOW] Reducing dimensionality via filter methods (correlations), wrapper methods (Recursive Feature Elimination), or tree-based feature importances.

Stage 6 — Machine Learning

Core statistical algorithms. Focus on model assumptions, mathematical operations, and hyperparameters.

1. Linear Regression (Regression)

[MUST KNOW] Foundational algorithm for continuous target prediction.

1. Intuition: Modeling a linear relationship between features and target by fitting a straight line that minimizes distance to the actual data points.

2. Mathematics: $y = w_0 + w_1x_1 + ... + w_nx_n + e$. Fits line by minimizing Mean Squared Error (MSE) via Ordinary Least Squares (OLS).

3. Assumptions: Linearity, Homoscedasticity (constant variance of residuals), Independence of errors, Normality of error distributions.

4. Hyperparameters: `fit_intercept` (bool).

5. Advantages: Simple, highly interpretable, fast training, no tuning needed.

6. Disadvantages: Assumes linear relations, sensitive to outliers and multicollinearity.

7. When to Use: Continuous target variable with clear linear relationships; baseline regression modeling.

8. When NOT to Use: Non-linear relationships or complex datasets with thousands of interacting features.

2. Logistic Regression (Classification)

[MUST KNOW] The foundational baseline for binary and multi-class classification.

1. Intuition: Predicts probability of membership in a class using the Sigmoid (logistic) function, outputting a value between 0 and 1.

2. Mathematics: $p = 1 / (1 + e^{-(\mathbf{w}^T\mathbf{x} + b)})$. Fits parameters via Maximum Likelihood Estimation.

3. Assumptions: Binary/Ordinal target, Independence of observations, Little or no multicollinearity among independent variables.

4. Hyperparameters: `C` (inverse regularization strength), `penalty` ('l1', 'l2', 'elasticnet'), `solver` ('lbfgs', 'saga').

5. Advantages: Extremely fast, outputs calibrated probabilities, easy to regularize, interpretable weights.

6. Disadvantages: Assumes linear decision boundary, struggles with complex relationships.

7. When to Use: Binary classification baselines; when model interpretability and probability outputs are critical.

8. When NOT to Use: Highly complex non-linear decision boundaries.

3. Decision Trees (Regression & Classification)

[MUST KNOW] Highly interpretable rule-based model.

1. Intuition: Making decisions by splitting data recursively on features that maximize information gain (Gini or Entropy reduction).

2. Mathematics: Splitting criteria: Gini Impurity = 1 - Sum(p_i^2); Entropy = -Sum($p_i * \log_2(p_i)$).

3. Assumptions: Non-parametric. No assumptions about data distribution.

4. Hyperparameters: `max_depth`, `min_samples_split`, `min_samples_leaf`, `criterion`.

5. Advantages: No feature scaling required, highly interpretable rules, handles non-linearities and missing values.

6. Disadvantages: High variance, extremely prone to overfitting, unstable (small data changes change the tree entirely).

7. When to Use: Simple rules-based modeling where interpretability is the priority.

8. When NOT to Use: When predicting smooth continuous variables, or when the decision boundary has diagonal elements.

4. Random Forest (Regression & Classification)

[MUST KNOW] Robust ensemble model using bagging of independent decision trees.

1. Intuition: Aggregates a large number of independent, deep decision trees trained on random bootstrap samples of data and random subsets of features.

2. Mathematics: Bootstrap Aggregating (Bagging) + Subspace Sampling. Combines independent trees by averaging (regression) or majority voting (classification) to reduce variance.

3. Assumptions: Non-parametric.

4. Hyperparameters: `n_estimators`, `max_features`, `max_depth`, `min_samples_leaf`, `bootstrap`.

5. Advantages: Very robust, avoids overfitting, handles high dimensionality, provides feature importance.

6. Disadvantages: Slower inference, large memory footprint, acts as a 'black box' compared to a single tree.

7. When to Use: General purpose classification and regression on tabular data.

8. When NOT to Use: Low latency real-time prediction environments, or when data is extremely sparse (e.g. text).

5. XGBoost (Regression & Classification)

[MUST KNOW] Highly optimized, industry-standard gradient boosting library.

1. Intuition: Sequential boosting of weak decision trees, where each tree is trained to correct the residual errors of the prior ensemble, utilizing L1/L2 regularization on leaf weights.

2. Mathematics: Taylor series expansion of objective: $Loss^{(t)}$ approx. Sum($g_i * f_t(x_i) + 0.5 * h_i * f_t^2(x_i)$) + penalty.

3. Assumptions: Non-parametric.

4. Hyperparameters: `eta` (learning_rate), `max_depth`, `lambda` (L2 reg), `alpha` (L1 reg), `subsample`, `colsample_bytree`.

5. Advantages: Native L1/L2 regularization, exceptional accuracy, handles missing values, parallel tree construction.

6. Disadvantages: Complex to tune, prone to overfitting if learning rate is too high, black box.

7. When to Use: Tabular datasets where maximum predictive accuracy is the main metric.

8. When NOT to Use: When interpretability is required by regulators, or when dataset is extremely small (use simpler models).

6. Principal Component Analysis (Unsupervised)

[MUST KNOW] Standard linear dimensionality reduction technique.

1. Intuition: Projects high-dimensional data onto orthogonal axes (principal components) that maximize the variance of the data, reducing dimensions with minimal information loss.

2. Mathematics: Computes the covariance matrix, then calculates its eigenvalues and eigenvectors. Eigenvectors corresponding to the largest eigenvalues represent principal components.

3. Assumptions: Linearity (variables have linear relations), Scaled data (highly sensitive to scale difference).

4. Hyperparameters: `n_components` (number of components or percentage of variance to keep).

5. Advantages: Removes multicollinearity, reduces memory footprint, speeds up model training.

6. Disadvantages: Principal components are linear combinations of features and lose direct interpretability.

7. When to Use: High dimensional datasets where features are highly correlated, and for preprocessing before modeling.

8. When NOT to Use: When maintaining individual feature interpretability is required, or when the data manifold is highly non-linear.

Stage 7 — Model Evaluation

Metrics selection determines model performance alignment with actual business KPIs.

Validation Strategy

[MUST KNOW] Train/Validation/Test split. Use Stratified K-Fold cross-validation for target class imbalances. Maintain temporal splits for timeseries.

Classification Metrics

[MUST KNOW] Accuracy (avoid on imbalanced sets), Precision (minimize false alarms), Recall (minimize missed detections), F1-Score (harmonic mean), ROC-AUC (overall ranking), PR-AUC (best for highly skewed classes).

Probability Calibration

[SHOULD KNOW] Ensuring predicted model probabilities match actual frequencies. Correct tree/SVM predictions via Platt Scaling or Isotonic Regression.

Bias-Variance Tradeoff

[MUST KNOW] Bias: error from underfitting (simple model). Variance: error from overfitting (complex model, fits training noise). Solve variance via regularization or bagging.

Stage 8 — Experimentation & A/B Testing

Formulate experiment structures to validate feature changes and quantify business impact.

A/B Test Design

[MUST KNOW] Control group (status quo) vs. Treatment group (feature variation). Establish randomization bounds to ensure users don't cross-contaminate.

Sample Sizing & MDE

[MUST KNOW] Determine sample size using: 1) Significance level (α, typically 0.05), 2) Statistical Power (1-β, typically 0.80), 3) Minimum Detectable Effect (MDE).

Hypothesis Validation & Pitfalls

[MUST KNOW] Calculate z-test or t-test significance on outcome KPIs. Pitfalls: Peeking (repeatedly checking p-values to stop test early, inflating Type I error), Multiple testing bias.

A/B Testing Business Scenario Example

Scenario: Product manager wants to test a new checkout button color. Conversion rate baseline is 5%. Target is to detect a 5% relative increase (MDE = 0.25 percentage points). With $\alpha = 0.05$ and Power = 0.80, sample sizing calculation dictates approximately 250,000 users per group. Split traffic evenly. After 2 weeks, check p-value: if $p < 0.05$, reject the null hypothesis and roll out the checkout button.

Stage 9 — Business & Product Analytics

Translating qualitative business operations questions into metrics, funnels, and structured analysis.

Business Metrics & KPIs

[MUST KNOW] LTV (Customer Lifetime Value), CAC (Customer Acquisition Cost), Churn Rate (user attrition), Retention Rate, ARPU (Average Revenue Per User).

Funnel & Cohort Analysis

[MUST KNOW] Funnel: tracking step-by-step drop-offs (e.g. Land page → Cart → Purchase). Cohort: tracking behaviors of groups sharing characteristics over time.

Translating Business to Analytics

[MUST KNOW] When asked 'Why did revenue drop?', translate to: 1) Decompose revenue = Active Users *Conversion Rate* Average Order Value. 2) Trace which component dropped. 3) Segment by traffic source or device.

Stage 10 — Time Series & Forecasting

Modelling data variables indexed chronologically.

Time Series Components

[MUST KNOW] Trend (long-term direction), Seasonality (repetitive cyclic patterns), and Noise (random variations).

Stationarity & Autocorrelation

[MUST KNOW] Stationary: constant mean/variance over time. Test using Augmented Dickey-Fuller (ADF) test. Autocorrelation (ACF) tracks correlation with past lags.

Forecasting Models

[SHOULD KNOW] ARIMA (p, d, q) and SARIMA (adds seasonal variables). Exponential Smoothing (Holt-Winters) models trend and seasonality exponentially.

Stage 11 — Advanced Data Science

Advanced modeling techniques for causal tracking, survivorship, and text analytics.

Causal Inference

[ADVANCED] Moving from correlation to causation. Focus on randomized controlled trials, matching methods, and difference-in-differences (DiD) designs.

Survival Analysis

[ADVANCED] Analyzing time-to-event data (e.g. time until customer churns). Focus on Kaplan-Meier curves and Cox Proportional Hazards models.

Recommendation Systems

[SHOULD KNOW] Collaborative Filtering (Matrix Factorization/SVD), content-based recommenders, and evaluation metrics (Precision@K, NDCG).

Natural Language Processing (NLP)

[SHOULD KNOW] Processing text data. Focus on TF-IDF word vectors, sentiment parsing, and transformer embeddings for textual feature columns.

Stage 12 — Data Science Engineering

ETL pipeline construction, Docker containers, API serving, and performance monitoring.

ETL/ELT Pipelines

[MUST KNOW] Extract-Transform-Load. Structuring automated pipelines to ingest database records, clean observations, and save model ready tables.

APIs & Containerization

[MUST KNOW] FastAPI/Flask to build endpoints exposing models. Docker containers to package dependencies and guarantee scaling stability.

Tracking & Monitoring

[SHOULD KNOW] Using MLflow or Weights & Biases for experiment tracking. Monitoring deployments for concept drift and performance decay.

Stage 13 — 20 Progression Projects

Building portfolio impact by resolving business-oriented and model accuracy problems.

Level 1 — Beginner Projects (5 Projects)

- 1. House Price Regressor (Boston/Ames dataset - Linear/Ridge, baseline MSE)
- 2. Iris Species Classifier (Iris dataset - Logistic Regression/KNN, multi-class metrics)
- 3. Titanic Survival Classifier (Titanic dataset - Decision Trees, categorical imputations)
- 4. Customer segmentation (K-Means, silhouette plotting on store user transactions)
- 5. Wine Quality Classifier (Wine dataset - Random Forest classifier, scaling comparison)

Level 2 — Intermediate Projects (5 Projects)

- 1. Credit Card Default Predictor (Kaggle - XGBoost/GBDT, class weight adjustments, PR-AUC)
- 2. E-Commerce Customer Cohort Analytics (Retail logs - Pandas cohort matrices, retention visualization)
- 3. Store Sales Time Series Forecaster (Store sales - ARIMA/SARIMA, lag feature engineering)
- 4. Movie Recommendation Engine (MovieLens - Matrix Factorization SVD, NDCG metrics)
- 5. Email Spam Classifier (Text emails - Naive Bayes + TF-IDF vectorizers)

Level 3 — Advanced Projects (5 Projects)

- 1. Real-Time Transaction Fraud Detector (FastAPI + Docker + LightGBM + Drift Monitoring)
- 2. Clinical Patient Survival Predictor (Cox Proportional Hazard, Kaplan-Meier curves)
- 3. Search Query Auto-Complete Engine (Trie trees index, prefix search lookups)
- 4. Dynamic Pricing Engine (Bayesian optimization, demand-based pricing)
- 5. Document Semantic Search Engine (BERT embeddings + Cosine similarity indexes)

Level 4 — Business & Product Projects (5 Projects)

- 1. A/B Testing Button Redesign Pipeline (MDE sizing, p-value calculations, power checks)
- 2. Customer Churn Economic Impact Modeller (Classifier + Revenue savings optimization matrices)
- 3. Marketing Attribution Modeller (Markov chains, attribution path conversions)
- 4. SaaS Product Funnel Optimization (Drop-off tracking, conversion optimization rates)
- 5. Store Inventory Allocation Optimization (Linear Programming, cost-demand modeling)

Stage 14 — Interview Preparation

Practicing case studies and data interpretation questions to demonstrate analytical strength.

Q: Python & SQL: Find User Retention

Write an SQL query to retrieve the percentage of users who logged in during Month 1 and returned in Month 2.

Q: Statistics: When does Central Limit Theorem fail?

CLT assumes independent and identically distributed (i.i.d.) variables with finite variance. It fails on heavy-tailed distributions (like Cauchy) with infinite variance.

Q: Probability: Describe Bayes Theorem diagnostic checks.

Used to calculate posterior probability of a disease given positive test: $P(D|+) = P(+|D)*P(D) / [P(+|D)*P(D) + P(+|\text{no } D)*P(\text{no } D)]$. Shows how base rates matter.

Q: ML Case Study: Design a ride-sharing ETA model.

Decompose into: 1) Business Problem (customer cancellation from inaccurate ETAs). 2) Features (hour of day, driver location, historical route speed). 3) Model (XGBoost Regressor). 4) Metrics (RMSE, MAPE).

Q: A/B Testing: How do you handle a flat A/B test?

If button conversion change is not significant ($p \geq 0.05$): 1) Accept Null Hypothesis. 2) Check segmentation (e.g. mobile vs. web, maybe it worked for a sub-demographic). 3) Conduct post-mortem check on sample size.

Recommended Timeline Plans & Checklists

Structured timetables designed for data science career progression.

Duration	Target Focus	Weekly Milestones
3-Month Plan	Core Analytics & SQL	Month 1: Python, SQL aggregations, Pandas joins. Month 2: Descriptive stats, probability, t-tests, ANOVA. Month 3: EDA visualizations and 5 Beginner projects.
6-Month Plan	ML & Evaluation	Months 1-3: Core analytics foundations. Month 4: Supervised ML (Linear/Logistic, XGBoost), PCA. Month 5: Model evaluations (ROC/PR-AUC), calibration. Month 6: 5 Intermediate projects.
12-Month Plan	Production & A/B Testing	Months 1-6: ML and analytics foundations. Month 7-8: A/B testing design, statistical power, MDE calculations. Month 9-10: Causal inference, survival, FastAPI pipelines. Month 11: 5 Advanced + 5 Business projects. Month 12: Case study preparation.

Final Data Science Specialist Checklist

[ ] Can explain the difference between t-test, ANOVA, and Chi-Square tests.

[ ] Can write SQL queries utilizing window ranking functions and consecutive session offsets.

[ ] Knows when to prioritize PR-AUC over ROC-AUC for imbalanced datasets.

[ ] Can design an A/B test, calculate sample size requirements, and check significance.

[ ] Understand the difference between MCAR, MAR, and MNAR missing data types.

[ ] Can decompose business metrics (e.g. churn) into clear features and analytical target columns.

[ ] Has implemented and documented at least one production-grade, API-deployed machine learning model.